

Master Plan Implementation Committee (MPIC) Solar Subcommittee Meeting Minutes Tuesday October 1, 2024

Committee Members Present: Teri Anderson, Susan Bronstein, Jen Milikowsky (via zoom), Doug Finn

School Department Present: Vito Perrone, School Superintendent

Consultants: Zachary Holt-PowerOptions, Inc.; Andreas Schmidt-SolectEnergy

Call to Order 10:10am

Presentation by Consultants:

Introduction: Zachary introduced PowerOptions – Based in Hopkinton, MA, a nonprofit energy buying consortium approved by MA Dept. of Capital Asset Management. Solect Energy was selected through competitive bidding process to be their solar designer/installer. They have done 800 commercial solar projects in MA including roof, ground, and canopy solar, batteries, and EV chargers. 100 public projects – schools, PVTAs, state, colleges, including Williamsburg and Buckland. Since Solect was competitively procured, the Town does not need to go out to bid. They have a network operations center in Hopkinton that monitors 750 projects.

Proposal:

- Andreas introduced the project preliminary design options and costs for elementary school. The Town can own the project directly or enter into a power purchase agreement (PPA) for PowerOptions to own and maintain the system and writes down the Town's energy costs at the school. It is recommended to plan a project for 2025 to take advantage of new SMART 3.0 program guidelines which will offer better arrangements for PPA agreements which could result in greater savings for the town. All agreements are reviewed by PowerOptions and DCAM.
- A PPA would be a fixed rate for 25 years.
- All staff are CORI checked and project installation can take place during school hours with minimal impact.
- If the town enters into a Letter of Intent, they can proceed with more detailed due diligence at no cost to the town including a site visit, a more detailed design, electrical engineering, and cost analysis. The only thing they can't do at this stage is geotechnical analysis - costs \$10-20,000 with a separate consultant which would need to be paid by the town upfront or by the grant. The budget with a grant assumes 20% contingency which would pay for worst case cost increases as a result of soil conditions. Silty/sandy soils = > expense for either deeper footings or spread footings. If town pays upfront, the PPA rate would be decreased based on our contribution to costs.
- They would apply in November for an interconnection agreement with EverSource for a 2025 install which would get that going and determine if there are any problems that need to be addressed.
- Agreed that we would adjust the pro formas to assume 100% of project costs covered by the MVP grant and the use of the 30% direct pay tax credit from the IRS. Depending on who owns the project – it would go to either PowerOptions or to the town. Need to

determine if that federal reimbursement would get subtracted from MVP grant (likely). It can take up to one year to get federal reimbursement so much plan on that.

- Need to select location for a canopy system
 - Back lot is adjacent to neighbors and we assume right at the property line setback. Need to confirm.
 - Canopy system has no roof, has ½ inch gaps between panels, is at a 30% pitch to drain and clean panels, drip line every 7 feet on long side. Have had no complaints with this design – no icing issues and no additional costs for sanding/salt, usually less plowing required.
 - Vito Perone still prefers a canopy system.
 - Yes, it is feasible and would be wise to maximize the size of canopy over the parking lot to anticipate any future increase in electric costs from new mini-splits or EV chargers. Will adjust the design.
 - Yes, there is 3-phase electric at the school but doesn't matter the location. It may be easier to get to the transformer at school via back lot, but either location is feasible.
 - Canopies are 20ft. tall. Ground mount is 8ft. tall with 6ft. chain link fence.
 - The canopy has a 14ft clear height sufficient for trucks and buses.
 - Footings are at 18-27ft intervals on outer edge. No additional drainage or loss of parking spaces is anticipated. System does not include water management. No roof, downspouts or drainage which would add 20-30% increased cost.
 - No advantage between systems for wind load safety. Both are designed for 90-110mph wind load required by code. High wind areas or hilltops require 130pmh load.
 - If we want battery backup, it is behind the meter and will need to separate critical loads and only is sufficient for 3 hours? Adds \$3-500K increased cost to the project! They don't recommend this. A generator is better and less costly.
 - They do not anticipate a constraint on the EverSource line for interconnection. It looks like the line is becoming saturated, but there is currently available kw to accommodate the school project, but we should apply soon.
 - Net metering is not the same as the interconnection agreement so the new MA DPU regulations re: net metering public cap exemptions for municipalities will not help with interconnection if the line needs to be upgraded. Interconnection Agreement (IA) takes 2-3 months to get approved.
 - We have one year to install the project after IA is approved. We should apply for the maximum size project anticipated because we can decrease the size and change the owner after approval but cannot increase the size.
 - The school has not completed all mini-split installation and is not using 30% in the gymnasium because it is not efficient and not warm enough in winter. So plan on at least 30% additional electric usage beyond current usage. Town is trying to get a green communities grant to do energy modeling for school to determine electric load.

Next Steps:

1. Need to set up another Solar Committee meeting soon to discuss how to proceed and to make a recommendation to Selectboard.
2. SolectEnergy needs to revised the pro forma after we tell them which option we prefer to size it for full parking lot coverage and to include the electric savings to the town which was inadvertently omitted from current version.

Submitted By,
Teri Anderson